

Authentication & Authorization

Task 1. Deep Conceptual Analysis (Otgonsuren Odmaa, Mungunbaatar Chingun)

Explain the full authentication flow in OAuth 2.0 and OpenID Connect.
Include: authorization code flow, tokens, refresh tokens, scopes, and identity verification.

Task 2. JWT Security Evaluation (Liu Hongjun, Cai Yupeng)

Analyze security problems of using JWT tokens in large-scale systems.
Discuss: token revocation, expiration strategy, signature attacks, and misconfiguration issues.

Task 3. Password Cracking Experiment (Mukhammadsodik, Ruziboy)

Write a small script that:

- Takes a hashed password (using SHA-256 or bcrypt)
 - Tries to brute-force a short list of common passwords
- Then write 8–10 sentences summarizing what this experiment teaches about password security.

Task 4. Role-Based Access Control (RBAC) Architecture (Batbileg Anujin, Erdenetogtokh Dulguun)

Design an RBAC model for a **University Management System**.
Include:

- Roles
 - Permissions
 - Hierarchy
 - Example authorization rules
- Diagram optional.

Task 5. Advanced Essay (Nasanbat Temuulen, Munkzolboo Tsolmon)

Write a **150–180-word** essay:
“Why Zero-Trust Authentication Models Are Essential in Modern Distributed Systems.”

Task 6. Attack Vector Analysis (Chan Myar Hnin Hsu, Ye Ming Aung)

Identify and explain **six advanced attack vectors** on authentication systems:
(Example: session fixation, replay attacks, CSRF, browser fingerprint spoofing).
Explain each in 3–4 sentences.

Task 7. Multi-Factor Authentication (MFA) Design (Salohiddin, Laziz)

Design an MFA system for a banking app that balances security and usability.
Discuss:

- Device binding
- Biometrics
- One-time codes
- Risk-based authentication
- Usability problems

Task 8. API Authorization (Alvin, Evans)

Explain how **fine-grained API authorization** works using:

- ABAC (Attribute-Based Access Control)
 - PBAC (Policy-Based Access Control)
- Compare the two and provide a complex scenario where each one is preferable.

Task 9. Security Audit Task (Eldric, Lim Xin Yu)

Perform a mock security audit of an authentication system and identify at least **7 vulnerabilities**.

For each vulnerability:

- Describe the risk
- Explain how attackers exploit it
- Provide a mitigation strategy